



Training Program 1: Advanced NLP & Large Language Models (LLMs)

- M01: Text Preprocessing
- M02: Word and Sentence Embeddings
- M03: Transformer Architectures
- M04: Entity-Centric NLP
- M05: Applied NLP Tasks
- M06: Multimodal NLP
- M07: LLM Fundamentals
- M08: Prompt Engineering
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- M10: RAG & Real-World Projects



Advanced NLP & Large Language Models (LLMs)

9.1 The Need for Efficient Fine-Tuning

9.2 LoRA: Low-Rank Adaptation Mechanics

9.3 QLoRA: Quantization for Parameter Efficiency

9.4 RLHF: Aligning Models with Human Feedback

9.5 Practical PEFT: A Hands-on Fine-Tuning Project

9.6 Summary and Next Steps

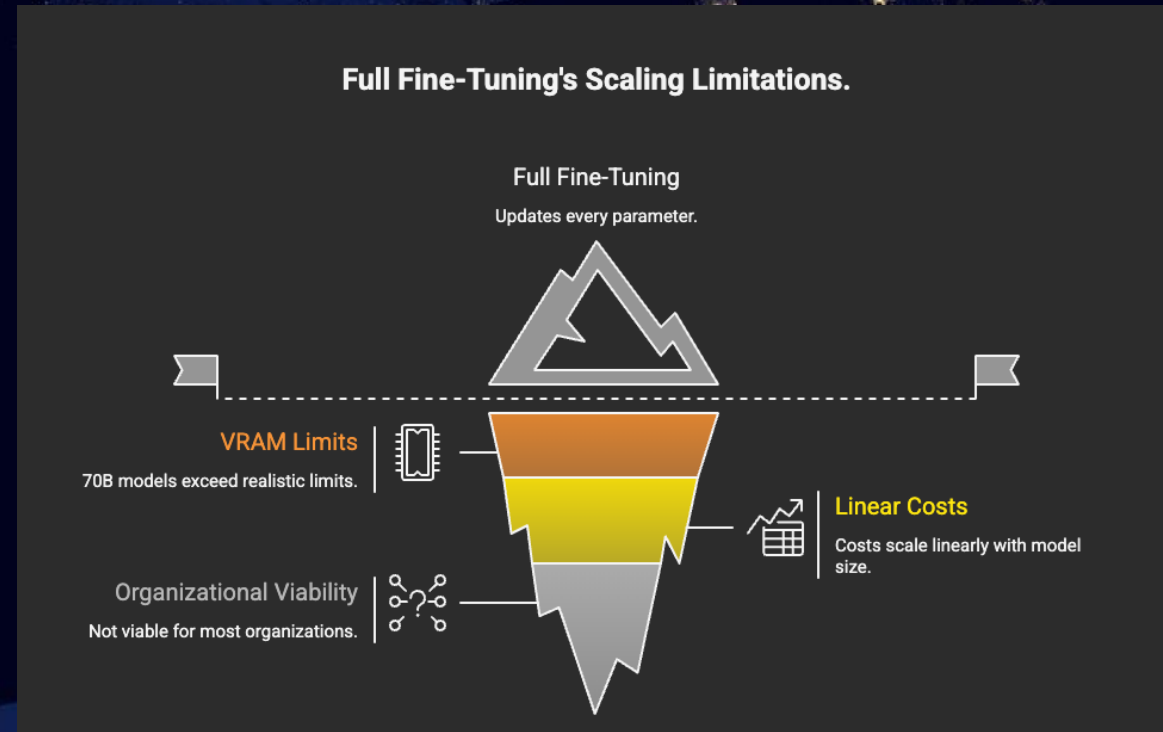
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Chapter 9.1 The Need for Efficient Fine-Tuning

Why Full Fine-Tuning No Longer Scales

Modern LLMs are too large to be fully fine-tuned in practical environments. Economic and hardware constraints force a shift toward efficiency-first adaptation strategies.

- Full fine-tuning updates every parameter
- 70B models exceed realistic VRAM limits
- Costs scale linearly with model size
- Not viable for most organizations



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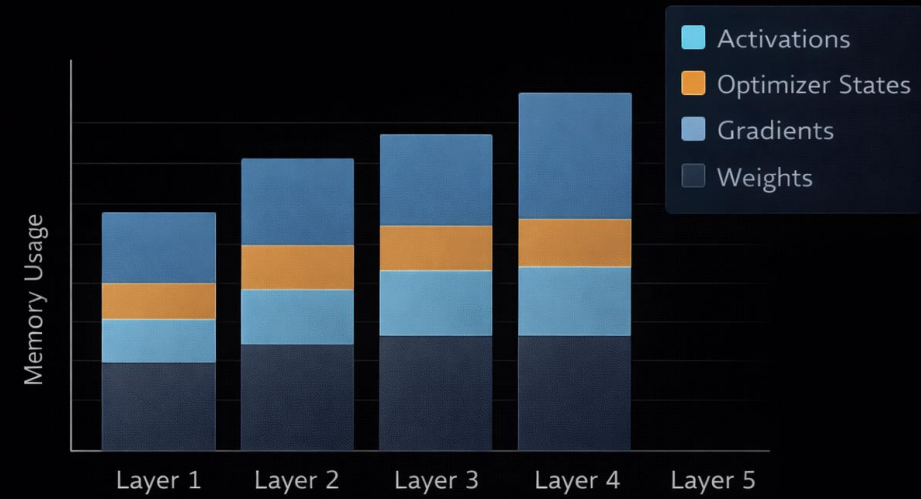
Chapter 9.1 The Need for Efficient Fine-Tuning

The VRAM Bottleneck

Training requires far more memory than inference due to gradients, optimizer states, and activations.

- ~16–20 bytes of VRAM per parameter
- 70B models require >1 TB VRAM
- Memory, not compute, is the main blocker
- Even multi-GPU setups struggle

GPU Memory Breakdown



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Chapter 9.1 The Need for Efficient Fine-Tuning

Catastrophic Forgetting

Aggressive fine-tuning on narrow datasets can erase general reasoning abilities.

- Model overwrites prior knowledge
- Loss of instruction-following skills
- Reduced generalization
- Especially risky with small datasets

Consequences of Catastrophic Forgetting

Overwriting Prior Knowledge

Loss of Instruction-Following

Reduced Generalization

Increased Risk with Small Datasets

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Chapter 9.1 The Need for Efficient Fine-Tuning

Why PEFT Became the Standard

PEFT freezes the base model and trains only a small number of new parameters.

- 90–95% VRAM reduction
- Preserves general intelligence
- Enables fast experimentation
- Ideal for domain adaptation





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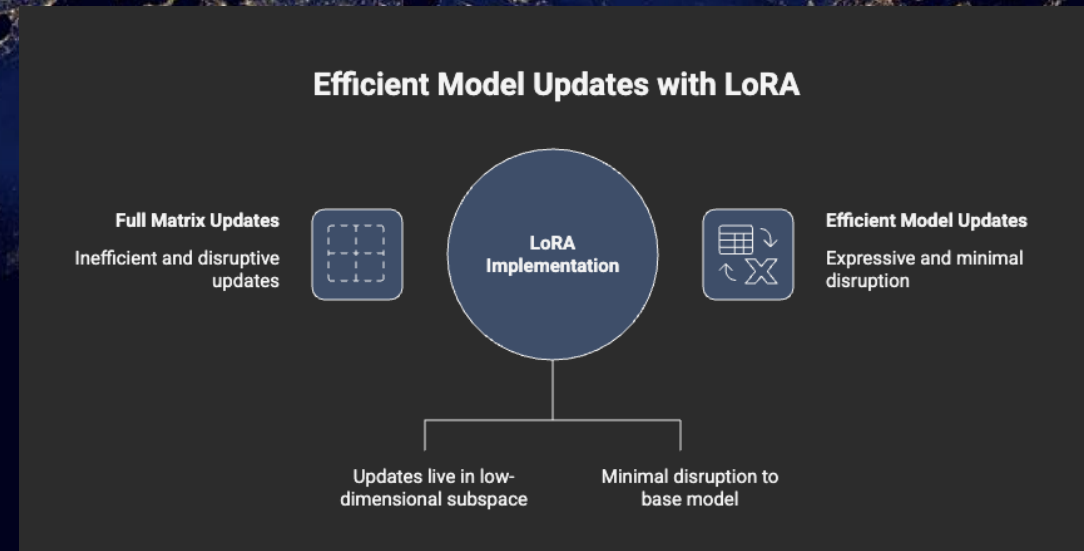
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Chapter 9.2 LoRA: Low-Rank Adaptation Mechanics

The Core Idea of LoRA

LoRA assumes task-specific updates live in a low-dimensional subspace.

- Weight updates are low-rank
- No need to modify full matrices
- Efficient and expressive
- Minimal disruption to base model



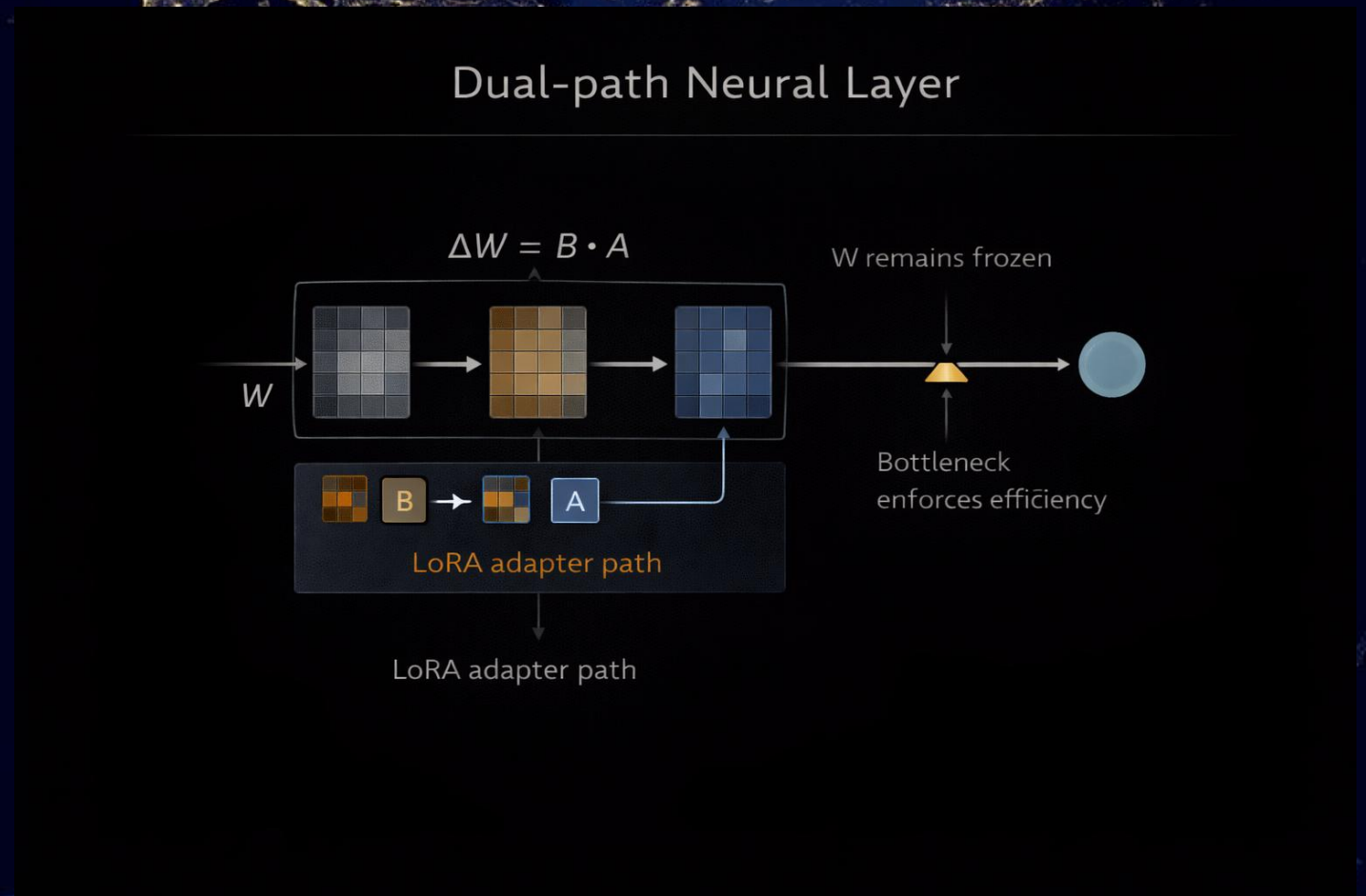
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Chapter 9.2 LoRA: Low-Rank Adaptation Mechanics

The Core Idea of LoRA

Instead of updating W , LoRA adds a parallel low-rank path.

- W remains frozen
- $\Delta W = B \cdot A$
- Bottleneck enforces efficiency
- Adapter path merged at inference



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Chapter 9.2 LoRA: Low-Rank Adaptation Mechanics

The Core Idea of LoRA

LoRA dramatically reduces training cost while preserving performance.

- 10,000× fewer trainable parameters
- Adapters are only megabytes
- Fast training and swapping
- Zero inference latency after merge

Optimizing LoRA Performance

Suboptimal LoRA Performance
Poor hyperparameter selection



Hyperparameter Tuning



Optimized LoRA Performance
Careful hyperparameter selection

Control LoRA capacity

Control update strength

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Chapter 9.2 LoRA: Low-Rank Adaptation Mechanics

Rank and Alpha Trade-offs

LoRA performance depends on careful hyperparameter selection.

- Rank controls capacity
- Alpha controls update strength
- Higher $\neq$ always better
- Typical r : 8–32

Comparison Chart: Full Fine-Tuning vs LoRA

	Full Fine-Tuning	LoRA
Parameters	All parameters	Low-rank updates
Storage	High storage cost	Low storage cost
Speed	 Slow	 Faster

LoRA adapter path

Adapter path merged at inference



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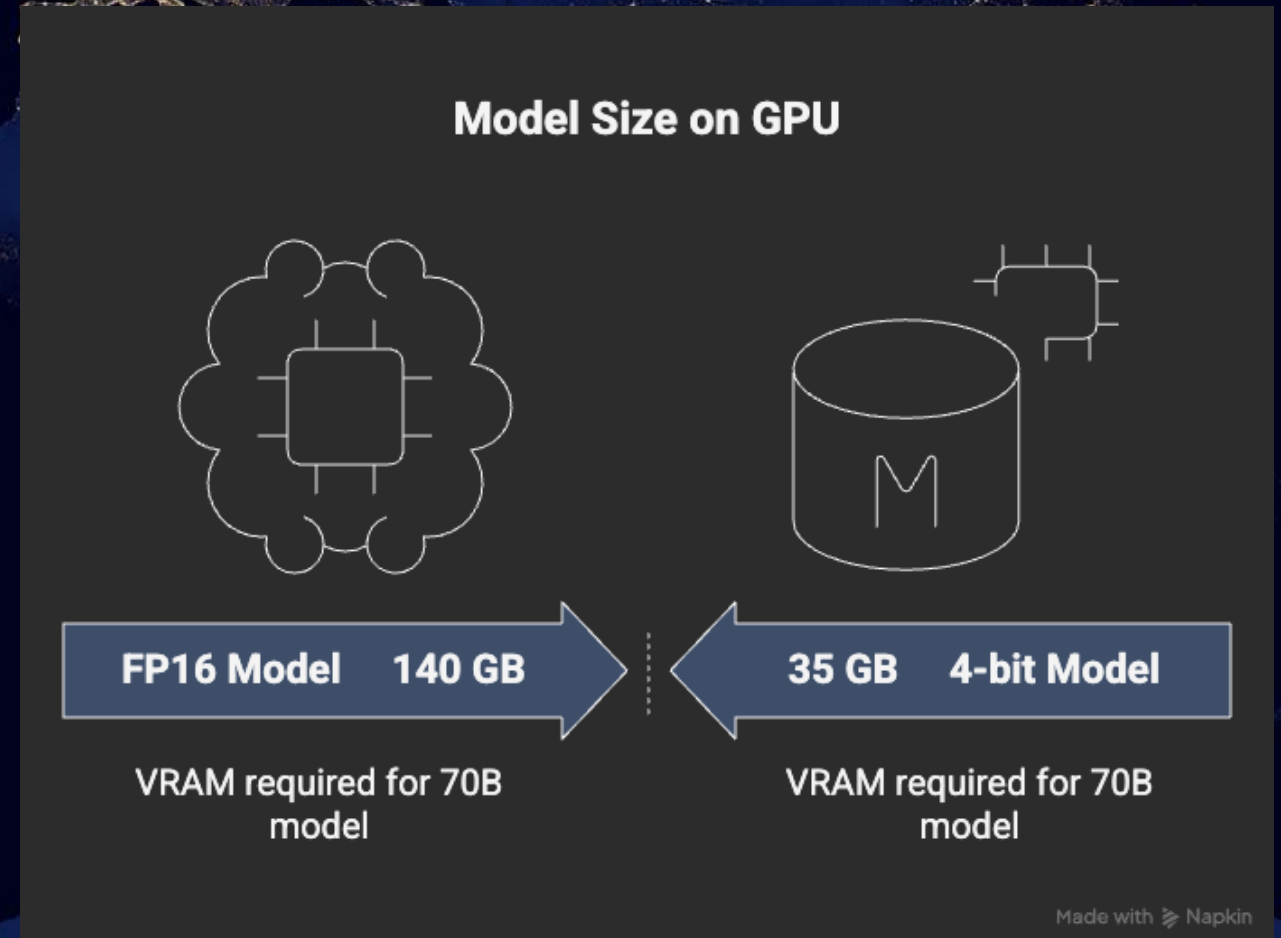
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Chapter 9.3 QLoRA: Quantization for Parameter Efficiency

Why Quantization Is Necessary

Even frozen models are too large to load in FP16.

- 70B FP16 $\approx$ 140 GB VRAM
- Consumer GPUs cannot load them
- Quantization solves the loading problem



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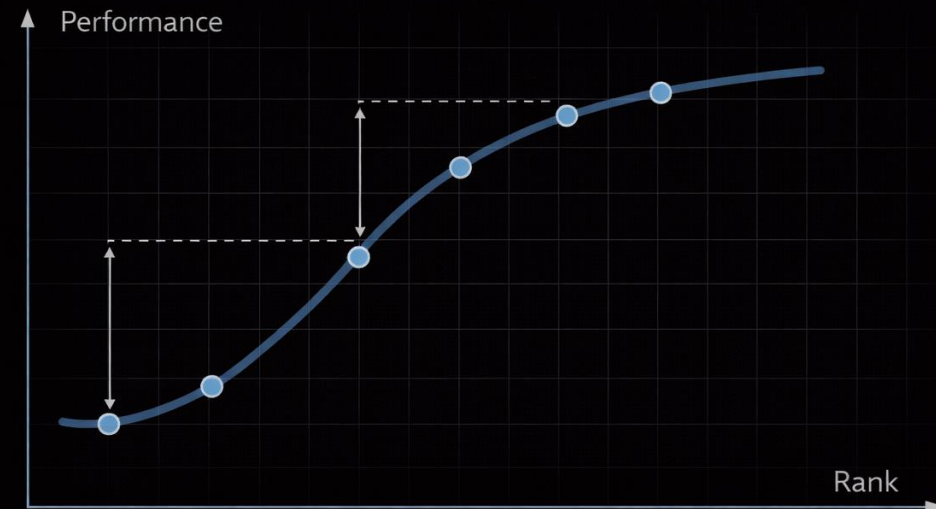
Chapter 9.3 QLoRA: Quantization for Parameter Efficiency

QLoRA: Training Through 4-bit Weights

QLoRA combines 4-bit quantization with LoRA adapters.

- Base model in NF4 (4-bit)
- LoRA trained in higher precision
- On-the-fly dequantization
- No accuracy collapse

Hyperparameter Tuning



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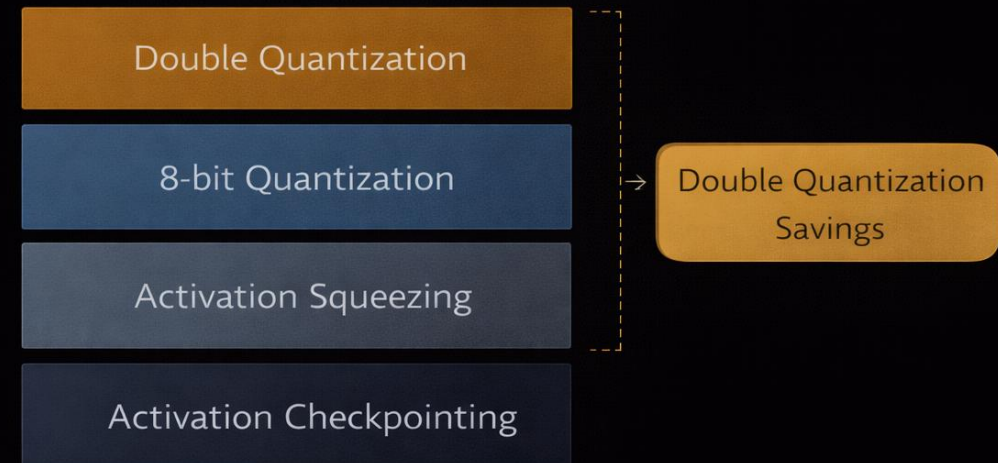
Chapter 9.3 QLoRA: Quantization for Parameter Efficiency

Double Quantization

QLoRA reduces memory even further by quantizing quantization constants.

- Saves several GB VRAM
- Critical for large models
- Enables single-GPU fine-tuning

Memory Optimization Layers





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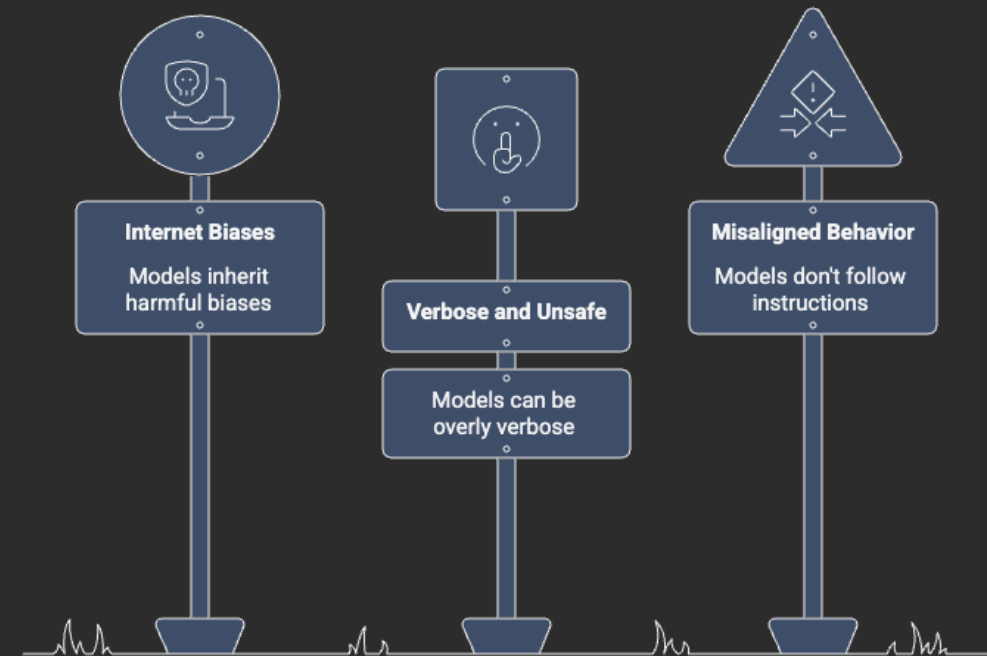
Chapter 9.4 RLHF: Aligning Models with Human Feedback

The Alignment Problem

Instruction-following alone does not guarantee safe or helpful behavior.

- Models inherit internet biases
- May be verbose, unsafe, or misaligned
- Alignment is mandatory for production

AI Alignment: A Critical Challenge



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Chapter 9.4 RLHF: Aligning Models with Human Feedback

The RLHF Pipeline

RLHF converts human preference into a learnable signal.

- Supervised Fine-Tuning (SFT)
- Reward Model training
- Reinforcement Learning (PPO)
- Human values → reward

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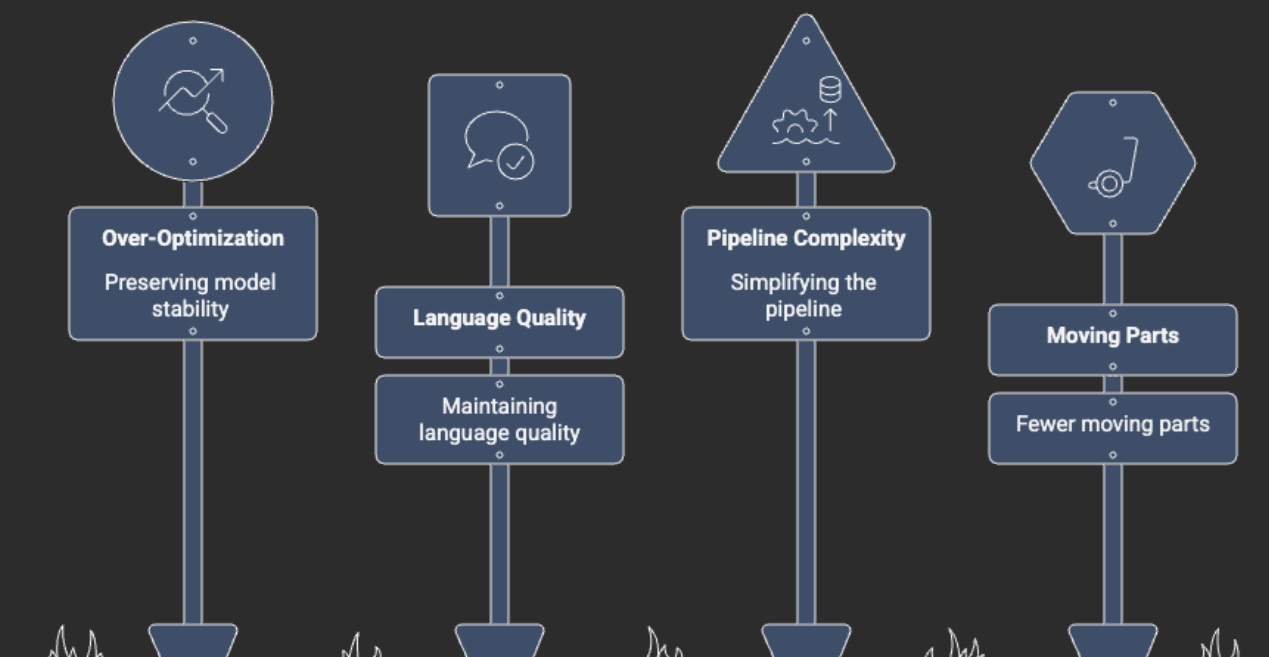
Chapter 9.4 RLHF: Aligning Models with Human Feedback

PPO and Modern Alternatives

PPO optimizes rewards while preserving model stability.

- Prevents over-optimization
- Maintains language quality
- DPO simplifies the pipeline
- Fewer moving parts

PPO and DPO: Balancing Rewards and Stability



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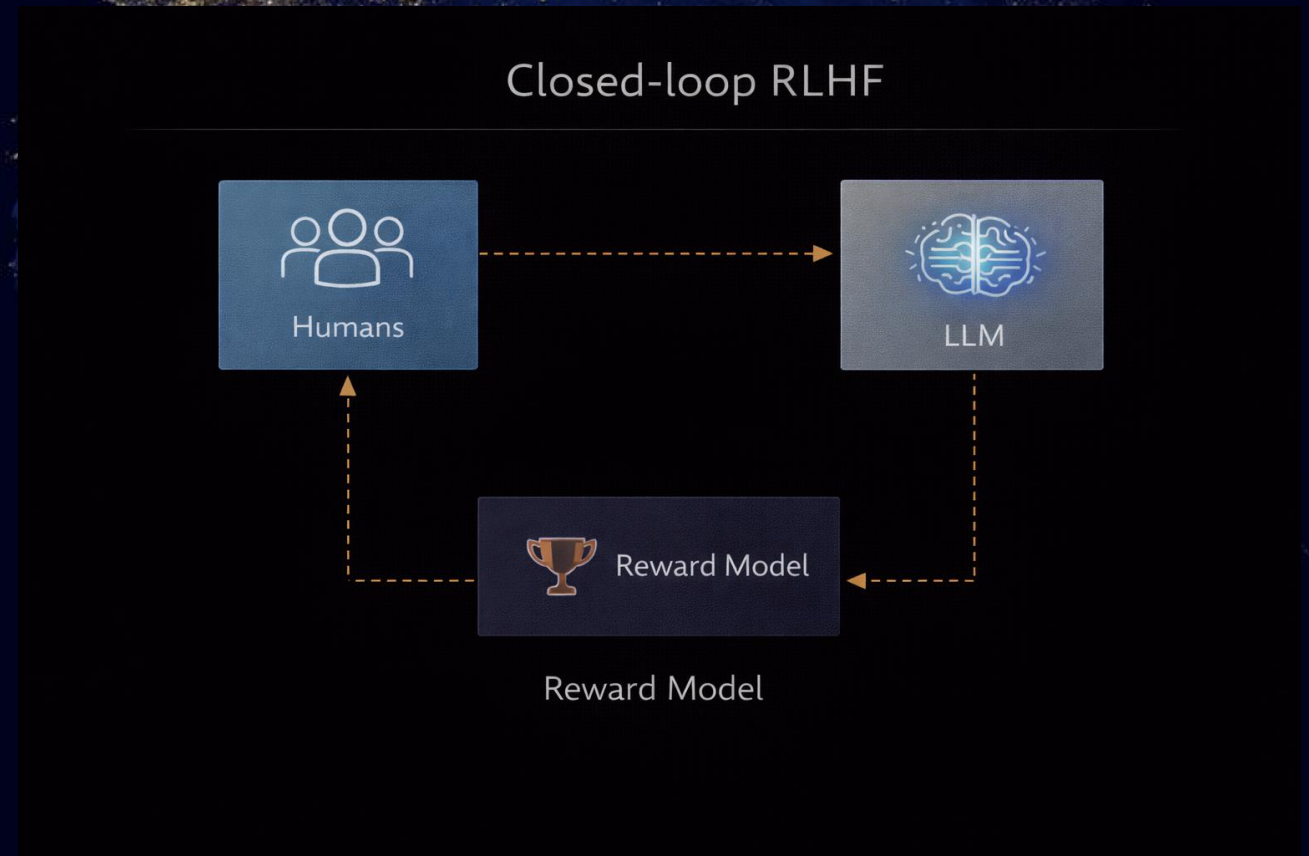
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Chapter 9.5 Practical PEFT: A Hands-on Fine-Tuning Project

Project Objective

The goal is to build a domain-expert assistant via PEFT.

- Legal audit assistant
- 2k–5k instruction-response pairs
- Strict data schema
- High-quality supervision



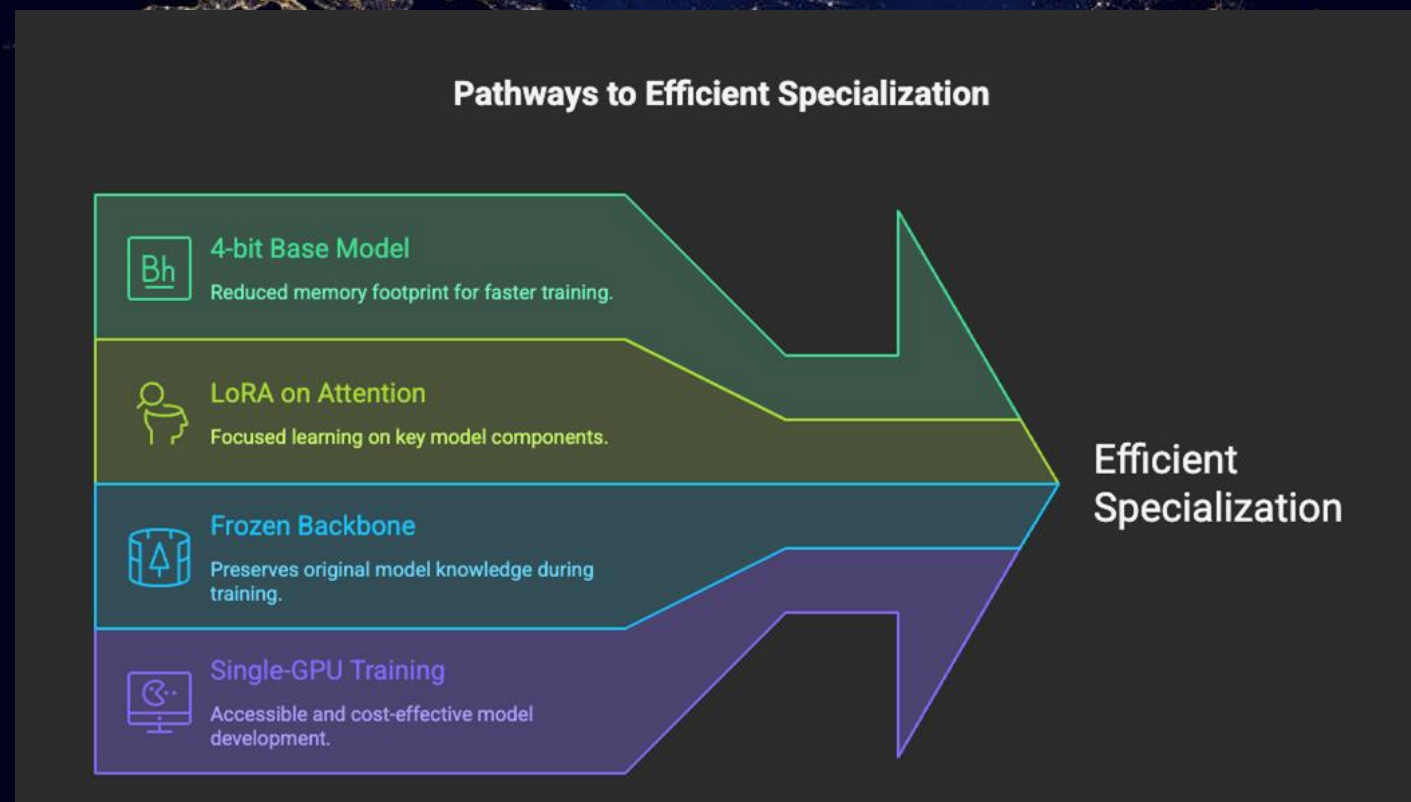
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Chapter 9.5 Practical PEFT: A Hands-on Fine-Tuning Project

Training Setup

QLoRA + LoRA enables efficient specialization.

- 4-bit base model
- LoRA on attention projections
- Frozen backbone
- Single-GPU training



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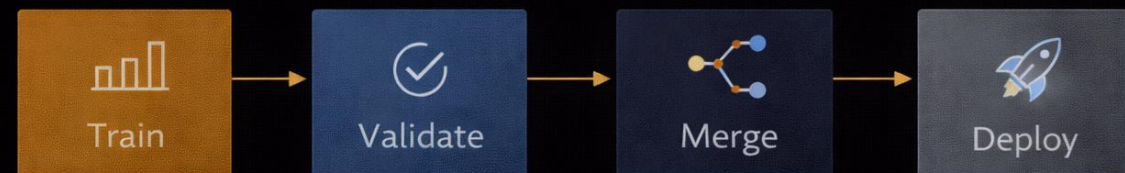
Chapter 9.5 Practical PEFT: A Hands-on Fine-Tuning Project

Evaluation and Deployment

Training is monitored and adapters are merged for deployment.

- Watch validation loss
- Test unseen samples
- Merge LoRA into base
- Produce single deployable model

Model Lifecycle





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Choosing the Right Adaptation Strategy

Fine-tuning must be treated as a strategic architectural decision, not as a default solution. Each adaptation technique introduces increasing levels of complexity, cost, and maintenance overhead. A mature AI system follows a layered escalation path, applying the minimum level of intervention required to achieve the desired performance and reliability.

- **Prompting first:** Use prompt engineering to shape behavior without modifying weights
- **RAG second:** Inject external knowledge before altering the model itself
- **PEFT when necessary:** Apply LoRA / QLoRA when style, structure, or domain expertise must be permanent
- **Full fine-tuning rarely justified:** Reserved for foundation-model creators or extreme specialization scenarios

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Chapter 9.6 Summary and Next Steps

The Future of PEFT

Model adaptation is evolving toward dynamic, modular intelligence, where specialization is no longer a static training event but a real-time, routable capability. Instead of maintaining multiple full models, future systems will rely on a single foundation model enhanced by a library of lightweight, interchangeable adapters.

- **Adapter routing:** Automatic selection of the most relevant LoRA at inference time
- **Mixture-of-Experts adapters:** Multiple specialized adapters activated in parallel
- **Instant specialization:** No retraining required per request
- **Shared backbone intelligence:** One model, many expert behaviors

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Key References

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